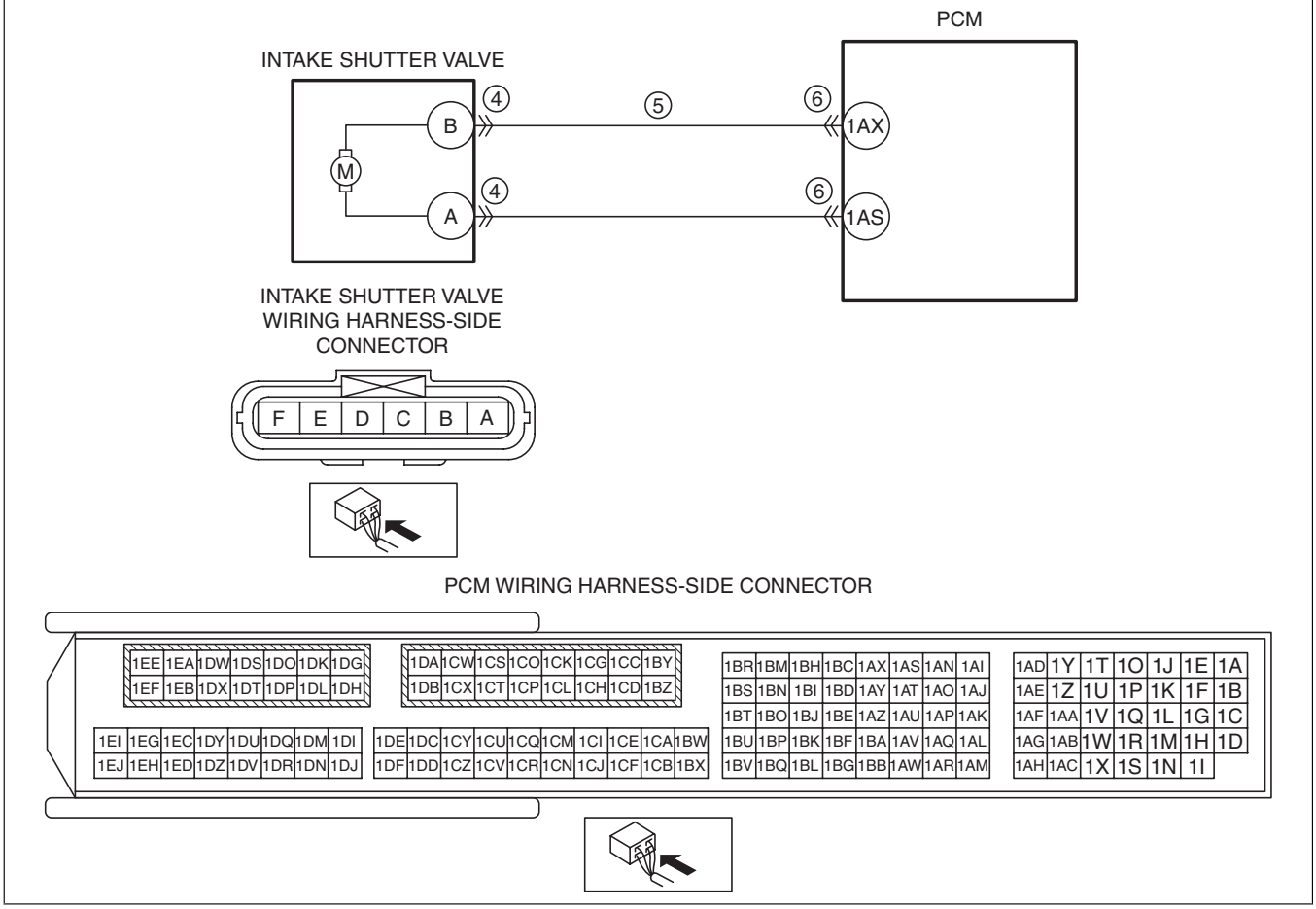
DTC P1589:00 [SKYACTIV-D 1.5]

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DTC P1589:00	Intake shutter valve circuit range/performance problem
DETECTION CONDITION	• The actual intake shutter valve opening angle is **4.2 °** larger than the target opening angle for a continuous **5 s** when the following conditions are met. — The following conditions continue for **2 s**: • Battery voltage: **8 V or more** • Target intake shutter valve opening angle: **below 75 °** • Target intake shutter valve opening angle is in the closed direction. • Change in speed of the target intake shutter valve: **Less than 0.25 °/msec**. — Intake shutter valve control: Being performed • The actual intake shutter valve opening angle is **4.2 °** smaller than the target opening angle for a continuous **5 s** when the following conditions are met. — The following conditions continue for **2 s**: • Battery voltage: **8 V or more** • Target intake shutter valve opening angle: **below 75 °** • Target intake shutter valve opening angle is in the open direction. • Change in speed of the target intake shutter valve: **Less than 0.25 °/msec**. — Intake shutter valve control: Being performed **Diagnostic support note** • This is a continuous monitor (CCM). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the auto DPF regeneration control. • Fully open the intake shutter valve. • Inhibits the compulsory DPF regeneration control. • Inhibits the EGR control. • PCM restricts engine-transaxle integration control. (ATX) • Inhibits engine-stop by operating the i-stop function.
POSSIBLE CAUSE	• Intake shutter valve connector or terminals malfunction • Short to ground in wiring harness between intake shutter valve terminal B and PCM terminal 1AX • PCM connector or terminals malfunction • PCM malfunction

DTC P1589:00	Intake shutter valve circuit range/performance problem



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS AT TIME OF DTC DETECTION TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture function of the PC. • Record the FREEZE FRAME DATA/snapshot data on the repair order.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**VERIFY RELATED PENDING CODE AND/OR DTC** • Switch the ignition off, then ON (engine off). • Perform the Pending Trouble Code Access Procedure and DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any other PENDING CODEs and/or DTCs present?	Yes	Go to the applicable PENDING CODE or DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.

STEP	INSPECTION		ACTION
4	**INSPECT INTAKE SHUTTER VALVE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the intake shutter valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 7.
		No	Go to the next step.
5	**INSPECT INTAKE SHUTTER VALVE CONTROL CIRCUIT FOR SHORT TO GROUND** • Verify that the intake shutter valve connector is disconnected. • Inspect for continuity between intake shutter valve terminal B (wiring harness-side) and body ground. • Is there continuity?	Yes	If the short to ground circuit could be detected in the wiring harness: • Refer to the wiring diagram and verify whether or not there is a common connector between intake shutter valve terminal B and PCM terminal 1AX. **If there is a common connector:** — Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. — Repair or replace the malfunctioning part. **If there is no common connector:** — Repair or replace the wiring harness which has a short to ground. If the short to ground circuit could not be detected in the wiring harness: • Replace the PCM (short to ground in the PCM internal circuit). (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to Step 7.
		No	Go to the next step.
6	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to the next step.
		No	Go to the next step.
7	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Drive the vehicle on a flat road surface 5 times under the following conditions. 1. Idle the engine for 20 s. 2. Shift to 2nd gear and rapidly accelerate the vehicle to **35 km/h {22 mph}** and drive the vehicle at a constant speed for **25 s**. 3. Stop the vehicle. 4. Switch the ignition to off and wait for **20 s or more**. 5. Shift to 3rd gear and rapidly accelerate the vehicle to **50 km/h {32 mph}** and drive the vehicle at a constant speed for **50 s**. 6. Idle the engine for **20 s**. • Perform the DTC Reading Procedure. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
8	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.